Energy materials and storage

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Of late, research on nanostructured materials has received considerable attention owing to their size and shape dependent properties that can be tailored for myriad applications in the fields of catalysis, medicine, optics and electronics. In particular, nanosized metal chalcogenides and metal oxides, as well as, their nanocomposites with diverse and unusual morphologies have aroused tremendous interest in view of their light driven catalytic functionalities, solar cells and batteries.

The present work is comprises the architecture of unique morphologies of the semiconductors, metal and ternary chalcogenides compounds using solvothermal methods. We have architectured a unique morphology of ZnO i.e. the hexagonal nanocups of ZnO with hexagonal cap using facile method. These ZnO nanocups have been used for variety of applications i.e. filed emission and photocatalyst. The nanodendrites of cobalt have been demonstrated using simple electrochemical method. The hexagonal cobalt nanoparticles, self-assembled and formed fern leave like morphology at different voltage. These dendrites have showed excellent antibacterial activity i.e. equivalent to silver. The ternary chalcogenides semiconductors have immense importance due to their layered structure such as ZnIn2S4 ternary chalcogenides semiconductor. The $ZnIn_2S_4$ marigold flowers, hollow structures have been demonstrated by solvothermal method. Interestingly, we obtained flowers with one micron cavity and these flowers are made up of 1-2nm thin nanopetals. These nanostructures are potentials in solar cells, batteries, high performance sensor and for biomedical applications.